Functions and arguments

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Functions and arguments before opening notes. Show any later correction separately.

Try the re**call** before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Lists and indexing

Without opening the old lesson, explain one key idea from Lists and indexing. Give an example and a mistake that the idea helps you avoid.

> **My remembered idea, example and correction**

Branching and boundary tests

Without opening the old lesson, explain one key idea from Branching and boundary tests. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Values variables and types

Without opening the old lesson, explain one key idea from Values variables and types. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

Complete the supported Functions and arguments trace and record the first step you can explain without help.

Use W2: Follow the execution order for **Function**s and **argument**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Functions and arguments explanation against the supplied facts before discussing it.

Use the worked case for **Function**s and **argument**s. Proposed explanation: “Writing def runs the body immediately”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Match positional arguments to parameters

> Predict both returned and printed values, showing which **argument** supplies first and second. What is displayed if the program contains only the **function definition** and no **call**s?
>
> Why does swapping these **argument**s change the result?

```
def difference(first, second):
    return first - second
print(difference(11, 4))
print(difference(4, 11))
```

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Label the parts

> Label the parts. Apply the supplied Functions and arguments facts and keep your reasoning.

In def show_remaining(capacity, used), identify the **function** name and **parameter**s. In show_remaining(15, 6), identify the **argument**s and their matching **parameter**s.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Follow the execution order

> Follow the execution order. Apply the supplied Functions and arguments facts and keep your reasoning.

Using the correct **definition**, predict show_remaining(8, 3), show_remaining(8, 8) and show_remaining(8, 0) in that order. Explain what happens if the file contains only the **definition**.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Repair ignored arguments

> Repair ignored arguments. Apply the supplied Functions and arguments facts and keep your reasoning.

The faulty body calculates 12 - 4 regardless of the **call**. State the **output** and expected **output** for show_remaining(11, 2). Repair the expression and name an earlier **call** to rerun.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Create a new function independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Create a new function independently. Apply the supplied Functions and arguments facts and keep your reasoning.

Each sealed box contains 6 counters. Define show_stock(boxes, loose) to print the total counters using its two **parameter**s. Assume nonnegative integer **argument**s. **Call** it with (3, 2), (0, 2) and (3, 0).

Predict each **output** and explain why show_stock(2, 3) is a different case.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 State the input assumption

> State the input assumption. Apply the supplied Functions and arguments facts and keep your

> reasoning.

The correct subtraction **function** prints -1 for show_remaining(4, 5). Does successful execution make this a valid cupboard record? Explain what the **function** assumes and what needs checking.

> **My reasoning and evidence**

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Functions and arguments to Python readiness portfolio, identify a limit and choose the next useful test.

Add a labelled **function**s and **argument**s **evidence** sheet to Python readiness portfolio. Show how you can distinguish a **function definition** from a **function call**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Functions and arguments

> Match arguments to parameters. A function prints the difference between capacity and reserved spaces. Explain the labelled step before checking the changed condition.

Match **argument**s to **parameter**s. A **function** prints the difference between capacity and reserved spaces.

Python 3: def show_free(capacity,reserved): print(capacity-reserved). **Call**s: show_free(11,3); show_free(3,11).

AI ACADEMY / PYTHON-BRIDGE/8

Functions and arguments

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Define without calling

The definition creates the function but prints nothing yet.

2. Match first arguments

capacity=11 and reserved=3 give a printed 8.

3. Trace reversed call

capacity=3 and reserved=11 give -8 under the same arithmetic.

CHANGE ONE THING

Remove both calls and leave the definition only.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Match every **argument** to a **parameter** for the two **call**s. Why do **output**s differ, and what prints when both **call**s are removed?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

Use actual Python 3 results to check the **prediction** after reasoning first.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Python readiness portfolio record.

Project: Python readiness portfolio

Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a **function** on changed **input**s.

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	

Evidence to keep	My record
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Match **argument**s to **parameter**s. A **function** prints the difference between capacity and reserved spaces.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? capacity=3 and reserved=11 give -8 under the same arithmetic.

1. Match first **argument**s

2. Trace reversed **call**

3. Define without **call**ing

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Trace reversed **call** capacity=3 and reserved=11 give -8 under the same arithmetic. Positional **argument**s match **parameter** names in order.

A **function definition** is not an execution of its body; separate **call**s supply separate **input** values.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Remove both **call**s and leave the **definition** only. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. No **output** is printed.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: No **output** is printed. The body

> runs only when the **function** is **call**ed.

Repair a missing building block

Review Lists and indexing (python-bridge/7). Goal: Distinguish a list index from the value stored at that position. Short practice: Label the three index positions.

Predict all original **output**s, then name the first **output** and stopping point after using index 3.. Return to this week and explain what you changed.

Review Branching and boundary tests (python-bridge/5). Goal: Trace the selected branch in an if elif else chain. Short practice: Trace -2, 4 and 5 through the branches.

Explain why the negative check comes first and what label -2 gets if it is removed.. Return to this week and explain what you changed.

Review Values variables and types (python-bridge/2). Goal: Distinguish integer, floating-point, text and Boolean values. Short practice: Predict both print **output**s and explain why they differ.

Does count+tag work, and what happens to each valid **output** after count becomes ten?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

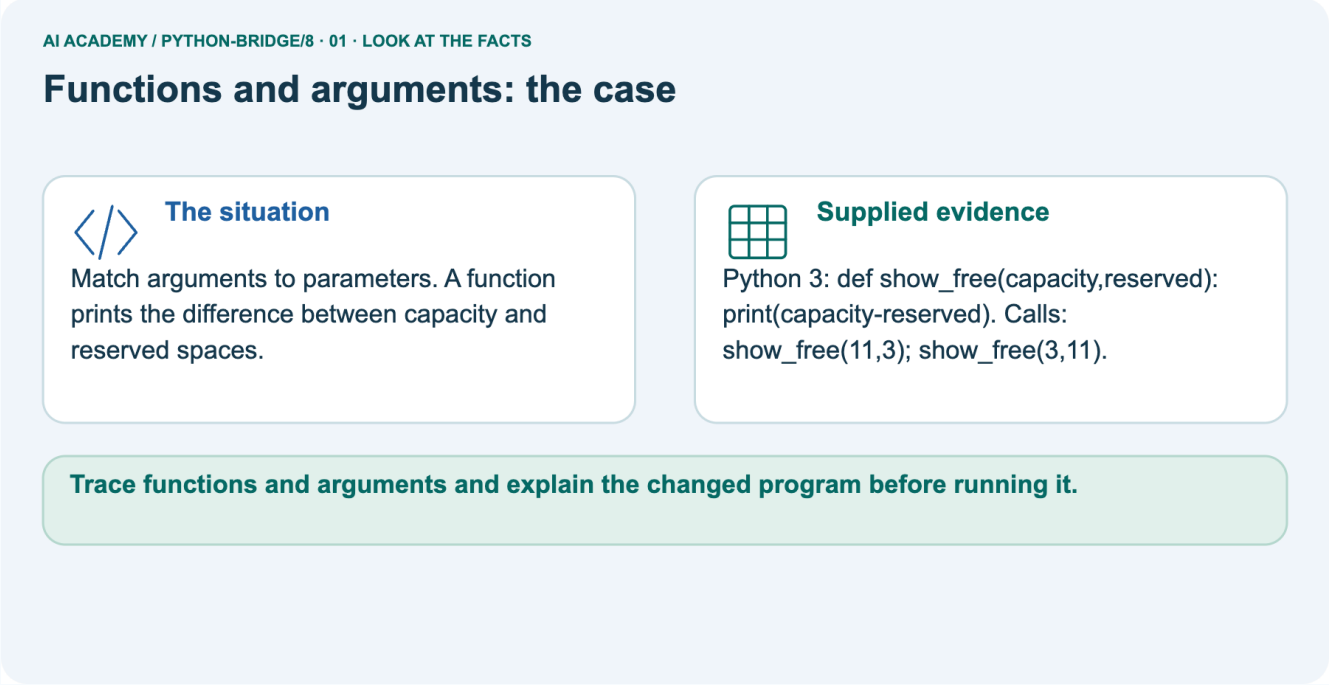
What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Functions and arguments picture from its labels, then record a relevant limit.

01. Functions and arguments: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

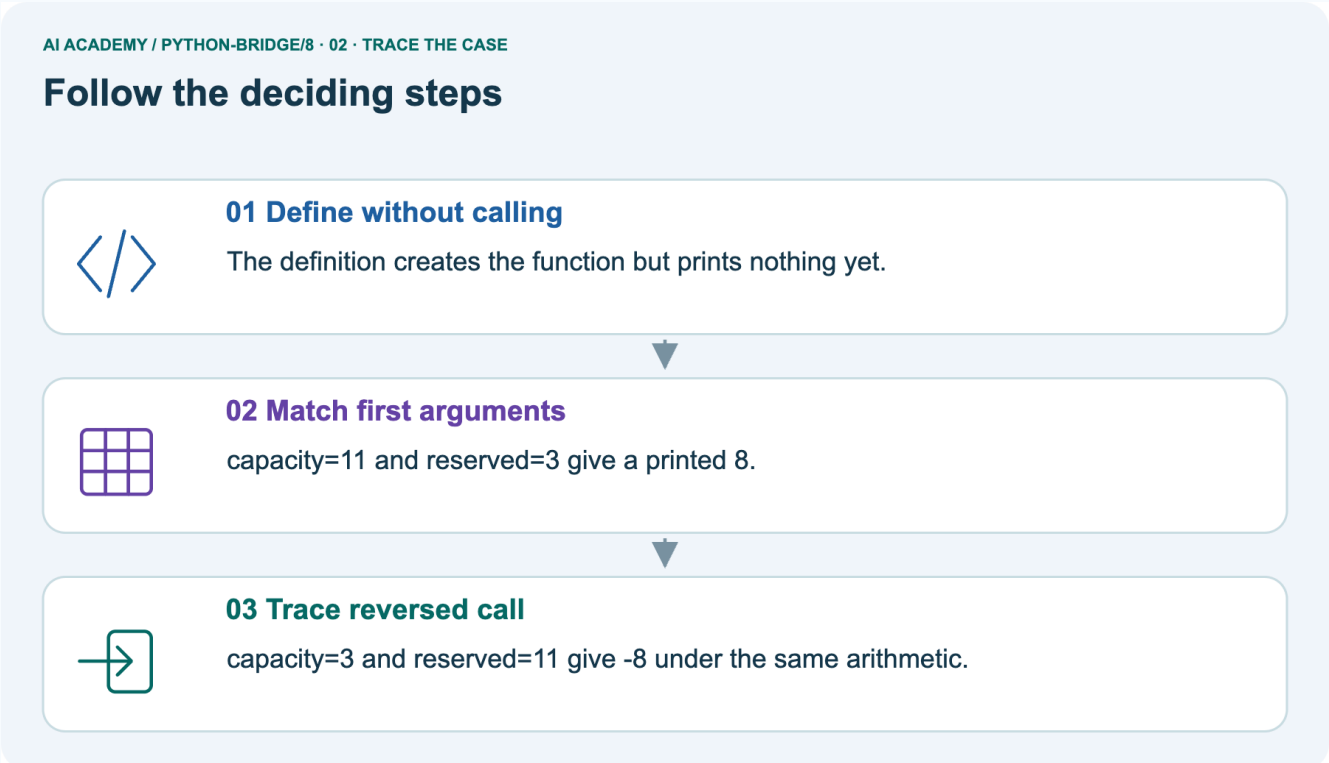
Match **argument**s to **parameter**s. A **function** prints the difference between capacity and reserved spaces. Supplied facts: Python 3: def show_free(capacity,reserved): print(capacity-reserved).

Calls: show_free(11,3); show_free(3,11).

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



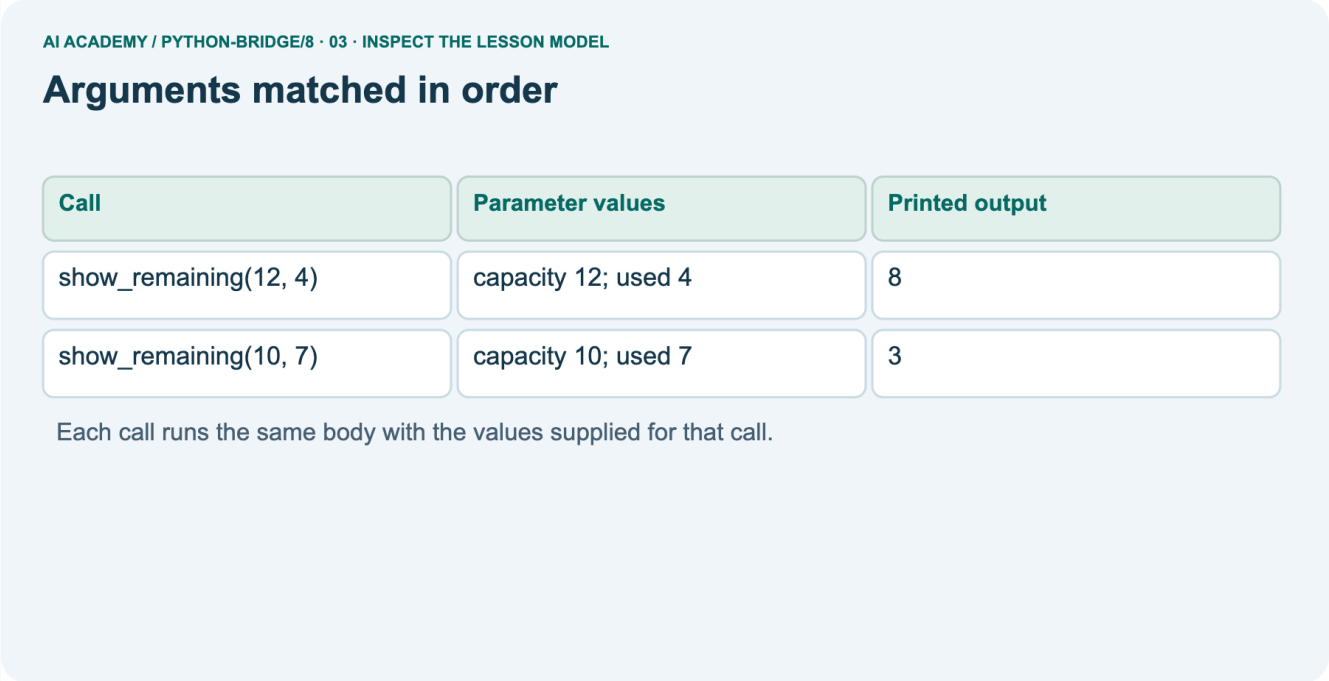
Follow the labelled picture with the matching explanation. Open the vector diagram

Define without **call**ing: The **definition** creates the **function** but prints nothing yet. → Match first **argument**s: capacity=11 and reserved=3 give a printed 8. → Trace reversed **call**: capacity=3 and reserved=11 give -8 under the same arithmetic.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Arguments matched in order



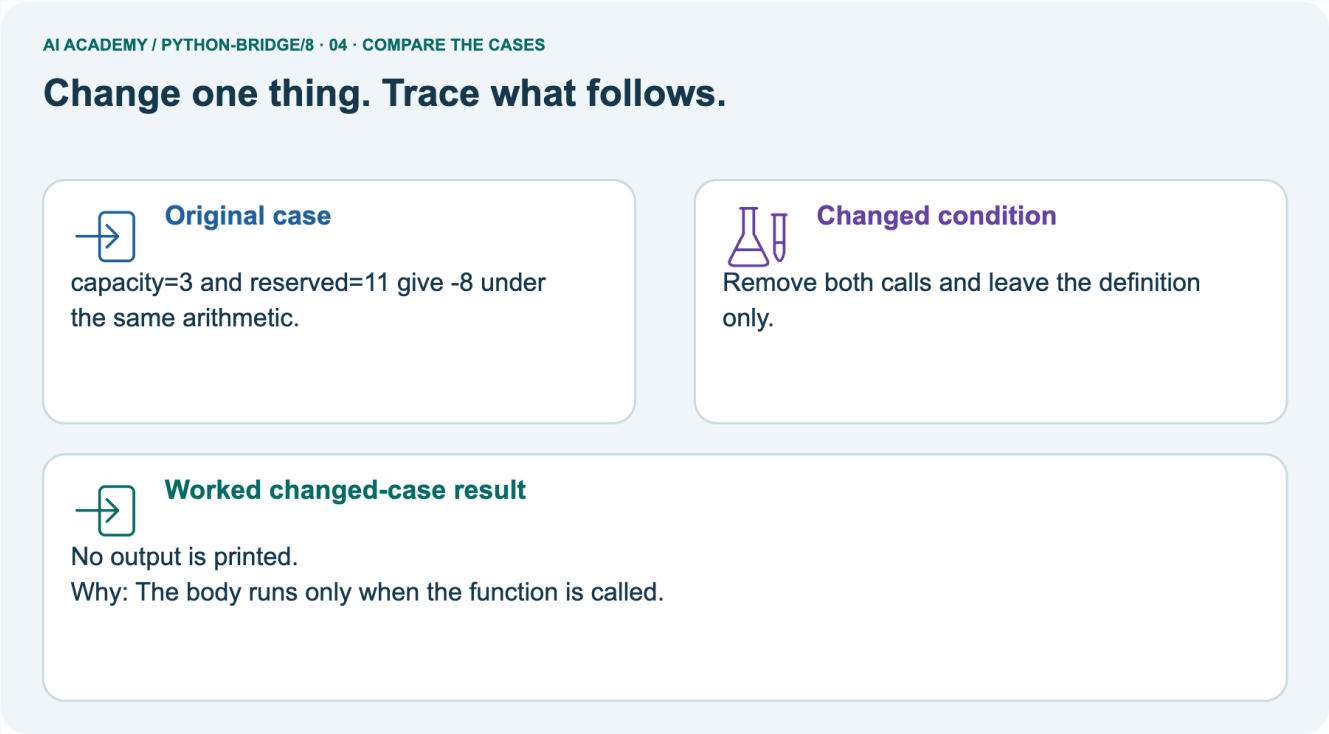
Follow the labelled picture with the matching explanation. Open the vector diagram

Arguments matched in order. show_remaining(12, 4) | capacity 12; used 4 | 8. show_remaining(10, 7) | capacity 10; used 7 | 3

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

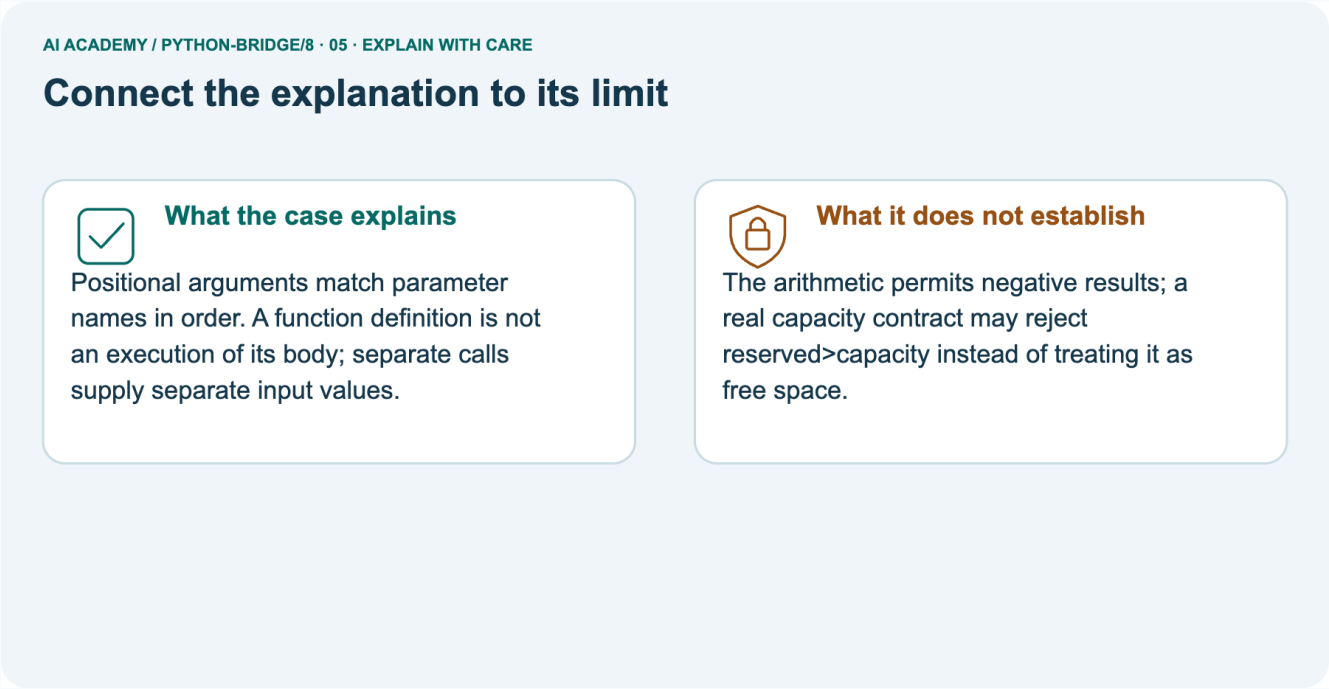
Original case: capacity=3 and reserved=11 give -8 under the same arithmetic. Changed condition: Remove both **call**s and leave the **definition** only. Expected result: No **output** is printed.

Reason: The body runs only when the **function** is **call**ed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

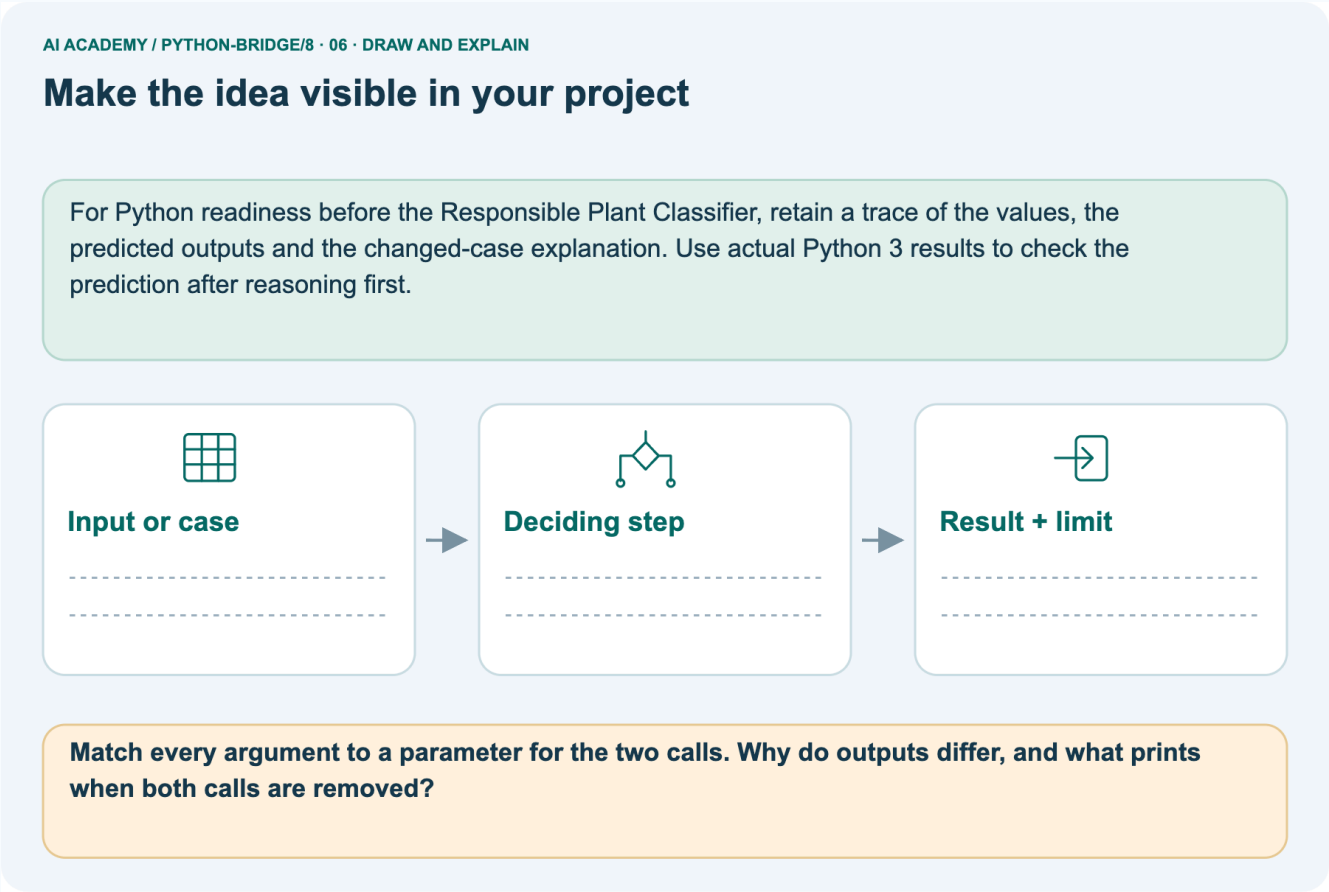
Positional **argument**s match **parameter** names in order. A **function definition** is not an execution of its body; separate **call**s supply separate **input** values.

Limit: The arithmetic permits negative results; a real capacity contract may reject reserved>capacity instead of treating it as free space.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Match every **argument** to a **parameter** for the two **call**s. Why do **output**s differ, and what prints when both **call**s are removed?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.